Matrices & Vectors

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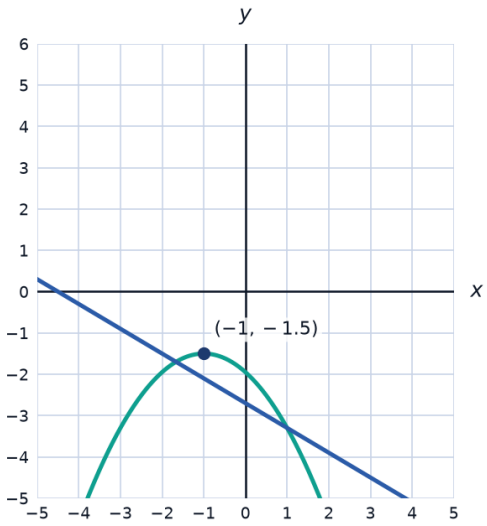
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Addition and scalar multiplication**

Let A be the matrix with rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$, and B be the matrix with rows $\langle -1, 4\rangle$ and $\langle 0, 2\rangle$. Find $A + B$ and $3A$.

2 **[Foundational]** • **Target: Addition and scalar multiplication**

Using the same A and B, find $A - 2B$.



3 **[Foundational]** • **Target: Matrix multiplication**

Using A (rows $\langle 2, 1\rangle$, $\langle 3, -1\rangle$) and B (rows $\langle -1, 4\rangle$, $\langle 0, 2\rangle$) from above, find AB.

4 **[Foundational]** • **Target: Matrix multiplication**

Let A have rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$, and let $\mathbf{v} = \langle 2, -3\rangle$. Find $A\mathbf{v}$.

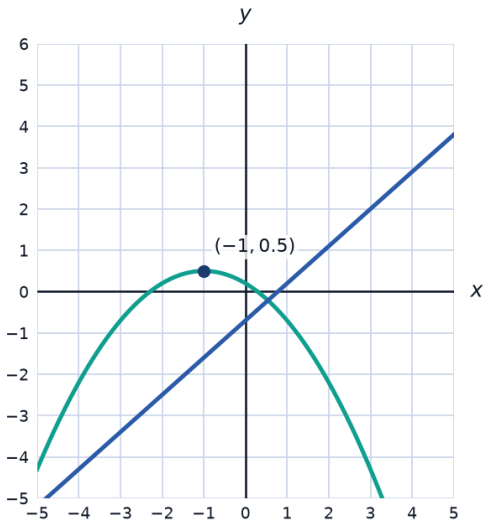
5 **[Foundational]** • **Target: Matrix multiplication**

Explain, without computing BA, why AB and BA need not be equal for 2×2 matrices.

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6 [Foundational] • Target: Determinant and inverse

Find $\det A$ for the matrix A with rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$, and state whether A is invertible.



7 [Foundational] • Target: Determinant and inverse

Find A^{-1} for the matrix A with rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Determinant and inverse

Without computing it, explain why the matrix with rows $\langle 4, 2\rangle$ and $\langle 6, 3\rangle$ has no inverse.

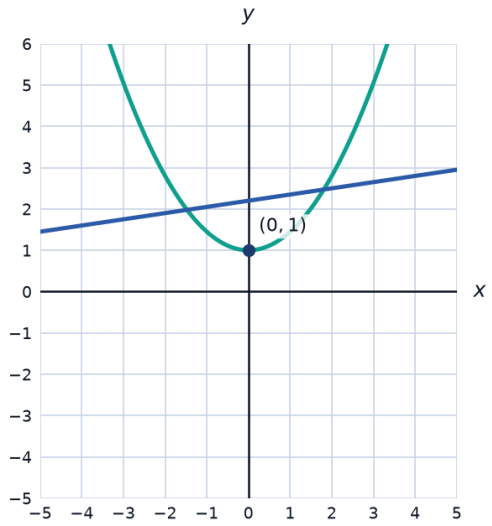
9 [Intermediate] • Target: Solving systems with the inverse matrix

Solve the system $3x - y = 8$ and $x + 2y = 5$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

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10 [Intermediate] • Target: Vector magnitude and dot product

What does $\mathbf{u} \cdot \mathbf{w} = 0$ tell you about $\mathbf{u}$ and $\mathbf{w}$ from the previous question?



11 [Intermediate] • Target: Angle between vectors

Find the angle between $\mathbf{p} = \langle 1, 0 \rangle$ and $\mathbf{q} = \langle 1, 1 \rangle$.

12 [Intermediate] • Target: Angle between vectors

Find the angle between $\langle 6, -8 \rangle$ and $\langle 3, 4 \rangle$, rounded to the nearest tenth of a degree.

13 [Intermediate] • Target: Addition and scalar multiplication

Given 3D vectors $\mathbf{u} = \langle 2, -3, 6 \rangle$ and $\mathbf{v} = \langle 1, 4, 2 \rangle$: (a) Find the unit vector in the direction of $\mathbf{u}$. (b) Calculate the angle θ between $\mathbf{u}$ and $\mathbf{v}$ to the nearest degree.

14 [Intermediate] • Target: Matrix multiplication

Find all real values of the constant k for which the matrix $M = (k + 2, 4; 3, k - 2)$ is singular (has no inverse).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

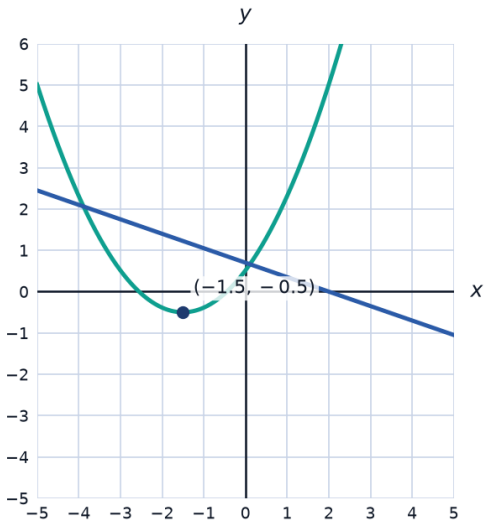
15 [Challenging] • Target: Determinant and inverse • Cross-Lesson: 1.01 Complex Numbers

Find the magnitude and direction angle θ (in degrees, $0^\circ \le \theta < 360^\circ$) of the vector $\mathbf{v} = \langle -5, 5\sqrt{3} \rangle$.

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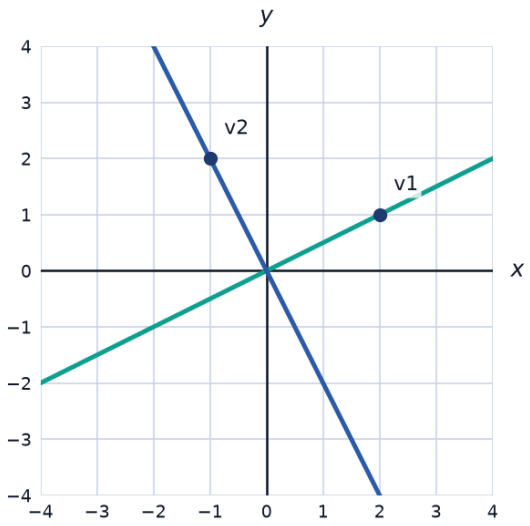
16 [Challenging] • Target: Solving systems with the inverse matrix • Cross-Lesson: 3.01 Sequences & Series

Find the angle θ between the vectors $\mathbf{a} = \langle 1, 2, 2 \rangle$ and $\mathbf{b} = \langle 2, -2, 1 \rangle$.



17 [Challenging] • Target: Vector magnitude and dot product • Cross-Lesson: 4.01 Advanced Functions & Equations

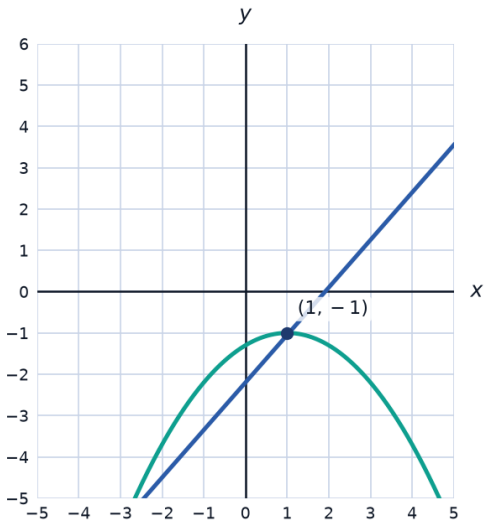
Find the inverse of the matrix $A = (4, 7; 1, 2)$.



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18 [Challenging] • Target: Angle between vectors • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Compute the matrix product AB where $A = (1, 3; -2, 4)$ and $B = (2, 0; 5, -1)$.



19 [Challenging] • Target: Addition and scalar multiplication • Cross-Lesson: 6.01 Limits

Vectors $\mathbf{a} = \langle 4, k \rangle$ and $\mathbf{b} = \langle 6, -8 \rangle$ are perpendicular. Find the value of k, and compute the projection of $\mathbf{c} = \langle 10, 5 \rangle$ onto $\mathbf{b}$.

20 [Challenging] • Target: Matrix multiplication • Cross-Lesson: 1.01 Complex Numbers

Let $A = (3, 1; 5, 2)$. Find A^{-1} and use it to solve the linear system: $3x + y = 7$ $5x + 2y = 12$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Addition and scalar multiplication

For $\mathbf{u} = \langle 20, 21 \rangle$ and $\mathbf{v} = \langle -5, -12 \rangle$, find $\cos\theta$ where θ is the angle between them, using $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

a. $\frac{352}{377}$

b. $-\frac{352}{377}$

c. $-\frac{27}{29}$

d. $-\frac{176}{21}$

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22 [Foundational] • Target: Matrix multiplication

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in the coordinate plane such that $|\mathbf{u}| = 5$, $|\mathbf{v}| = 10$, and the angle θ between them is obtuse ($90^\circ < \theta < 180^\circ$). If the minimum value of $|t\mathbf{u} + \mathbf{v}|$ over all real numbers t is equal to 6, what is the value of the dot product $\mathbf{u} \cdot \mathbf{v}$?

a. -50

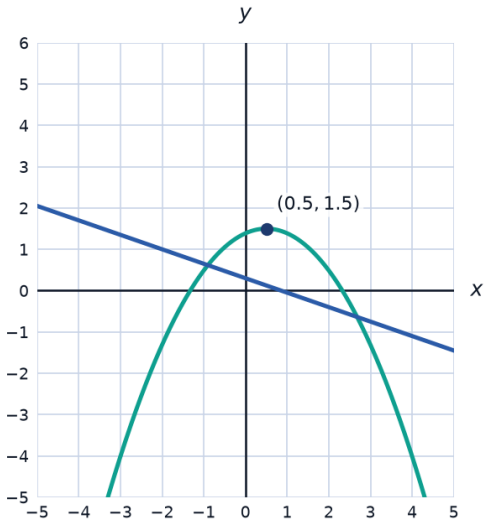
b. -40

c. -30

d. 40

23 [Foundational] • Target: Determinant and inverse

In the standard (x, y)-coordinate plane, vectors $\mathbf{u} = \langle a, b \rangle$ and $\mathbf{v} = \langle c, d \rangle$ have magnitudes $||\mathbf{u}|| = 13$ and $||\mathbf{v}|| = 5$. If their dot product is $\mathbf{u} \cdot \mathbf{v} = 25$, and the matrix $M = [a, b; c, d]$ has $\det(M) > 0$, what is the value of $\det(M)$?



a. 25

b. 40

c. 60

d. 65

24 [Foundational] • Target: Solving systems with the inverse matrix

For all real values of θ, the matrix product $[\cos\theta, \sin\theta][6, 2; 2, 3][\cos\theta; \sin\theta]$ evaluates to a 1×1 matrix $[k]$. What is the maximum possible value of k?

a. 6

b. 7

c. 9

d. 14

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25 [Foundational] • Target: Vector magnitude and dot product

Matrices A and B are defined as $A = [2, k; -1, 3]$ and $B = [1, 4; 2, k]$, where k is a real number. If the determinant of the matrix sum $A + B$ is equal to 17, what is the value of k? (Note: The determinant of a 2×2 matrix $[a, b; c, d]$ is defined as $ad - bc$.)

a. 1

b. 2

c. 6

d. $\frac{19}{2}$

26 [Foundational] • Target: Angle between vectors

For a 2×2 matrix $[a, b; c, d]$, the determinant is defined as $ad - bc$. Given the matrices $A = [x, 2; -1, 3]$ and $B = [1, -4; 3, x]$, the determinant of the matrix $A + B$ is equal to 19. If $x > 0$, what is the value of x?

a. 2

b. $\frac{5}{4}$

c. 4

d. 6

27 [Foundational] • Target: Addition and scalar multiplication

In the standard coordinate plane, vector $\mathbf{u} = \langle 2, -1 \rangle$ and vector $\mathbf{v} = \langle k, 5 \rangle$, where k is a real constant. If the magnitude of the vector $\mathbf{u} + \mathbf{v}$ is equal to 5, which of the following is a possible value of k?

a. -5

b. -1

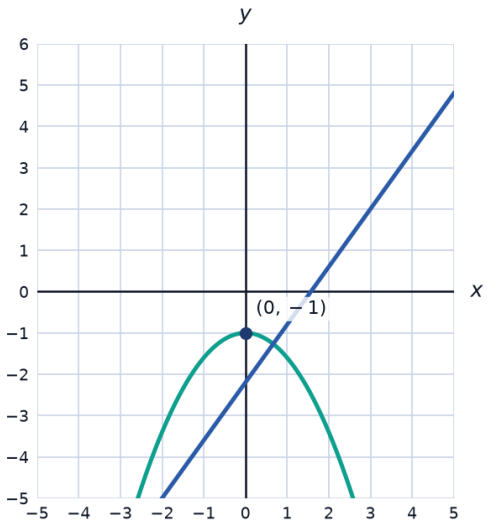
c. 3

d. 7

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28 [Foundational] • Target: Matrix multiplication

What is the value of the determinant of the matrix $[7, -3; 2, 4]$?



a. 34

b. 22

c. -29

d. -34

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Determinant and inverse

Find the angle θ between $\mathbf{u} = \langle -6, 0 \rangle$ and $\mathbf{v} = \langle 6, -1 \rangle$, to the nearest 0.1°.

a. 170.5°

b. 9.5°

c. 180.5°

d. 165.5°

30 [Intermediate] • Target: Solving systems with the inverse matrix

Solve using the inverse-matrix method: $\{-5x + 3y = 28;\ -5x + 0y = 25$.

a. x=-5,\ y=1

b. x=1,\ y=-5

c. x=-4,\ y=1

d. x=-5,\ y=-1

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31 [Intermediate] • Target: Vector magnitude and dot product

Given the vectors $\mathbf{u} = \langle 3, -4 \rangle$ and $\mathbf{v} = \langle -2, 5 \rangle$, what is the vector $2\mathbf{u} + \mathbf{v}$?

a. $\langle 4, -3 \rangle$

b. $\langle 1, 1 \rangle$

c. $\langle 4, 1 \rangle$

d. $\langle 8, -13 \rangle$

32 [Intermediate] • Target: Angle between vectors

If matrix $A = [3, -2; 0, 4]$ and matrix $B = [-1, 5; 2, -3]$, what is the matrix $2A + B$?

a. $[5, 1; 2, 5]$

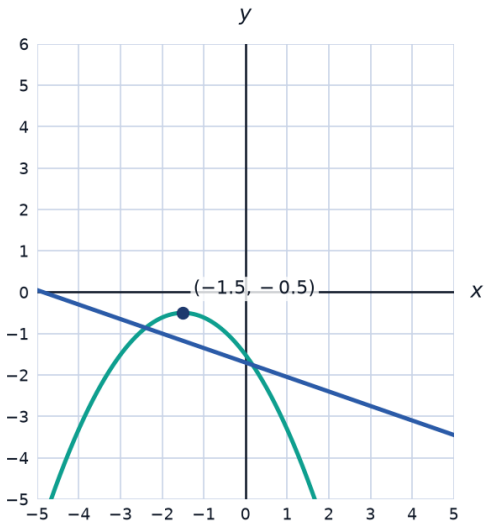
b. $[2, 3; 2, 1]$

c. $[5, 3; 2, 1]$

d. $[7, -9; -2, 11]$

33 [Intermediate] • Target: Addition and scalar multiplication

Given matrix $A = (2, -1; 3, 3)$, what is the scalar product $3A$?



a. $(5, 2; 6, 6)$

b. $(6, -1; 3, 9)$

c. $(6, -3; 9, 9)$

d. $(6, 3; 9, 10)$

34 [Intermediate] • Target: Matrix multiplication

A transformation maps point $A(1, 2)$ to $A'(-4, 2)$ and point $B(3, 5)$ to $B'(-2, 5)$. This transformation can be represented by matrix multiplication $(x\ y)' = M(x\ y) + (a\ b)$. Find the values of a and b.

a. a = -5, b = 0

b. a = 5, b = 0

c. a = 0, b = -5

d. a = 0, b = 5

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35 [Intermediate] • Target: Determinant and inverse

Consider a linear transformation T represented by the matrix $A = (2, 1\ 1, 3)$. Let $P = (x; y)$ be a point in the plane. The transformation maps P to P'. If the area of the region transformed by T from a unit square in the xy-plane is A, and the determinant of A is $\det(A)$, find the value of $\det(A)$ if the area of the transformed region is 5.

a. 5

b. 2

c. 1

d. 10

36 [Intermediate] • Target: Solving systems with the inverse matrix

Consider the matrix $A = (2, 0; 0, 3)$. When the matrix A is applied to the vector $\mathbf{v} = (x; y)$, the resulting vector is $A\mathbf{v} = (6; 9)$. What is the original vector $\mathbf{v}$?

a. $(3; 3)$

b. $(1; 2)$

c. $(6; 3)$

d. $(12; 27)$

37 [Intermediate] • Target: Vector magnitude and dot product

Compute the dot product $\mathbf{u} \cdot \mathbf{v}$ for $\mathbf{u} = \langle 2, 3 \rangle$ and $\mathbf{v} = \langle 2, 5 \rangle$.

a. 19

b. 22

c. 16

d. 14

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Angle between vectors • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

What is the magnitude of the 2D vector $\mathbf{v} = \langle 3, -4 \rangle$?

a. 5

b. 12

c. 10

d. 7

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39 [Challenging] • Target: Addition and scalar multiplication • Cross-Lesson: 6.01 Limits

In ΔABC, the side lengths are $AB = 5$ and $BC = 4$, and $\angle B$ is an acute angle. If $\cos(2B) = -\frac{31}{32}$, what is the length of side AC?

a. 6

b. $\sqrt{46}$

c. $\sqrt{41}$

d. $\sqrt{31}$

40 [Challenging] • Target: Matrix multiplication • Cross-Lesson: 1.01 Complex Numbers

In ΔABC, the lengths of sides AB and BC are 5 and 8, respectively, and the measure of $\angle B$ is 60°. What is the length of the altitude from vertex B to side AC?

a. $\frac{20\sqrt{3}}{7}$

b. $\frac{40\sqrt{3}}{7}$

c. $\frac{10\sqrt{3}}{7}$

d. $\frac{20}{7}$

41 [Challenging] • Target: Determinant and inverse • Cross-Lesson: 3.01 Sequences & Series

In ΔABC, the length of side AB is x, the length of side BC is $x + 3$, and the length of side AC is 7. If $\angle B = 60^\circ$, what is the value of x?

a. 5

b. 8

c. 4

d. 6

42 [Challenging] • Target: Solving systems with the inverse matrix • Cross-Lesson: 4.01 Advanced Functions & Equations

In ΔABC, $A = 64^\circ$, $a = 18$, $b = 9$. Find angle B to the nearest tenth (assume B acute).

a. 26.7°

b. 28.7°

c. 89.3°

d. 23.7°

43 [Challenging] • Target: Vector magnitude and dot product • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

In ΔABC, $a = 13$, $b = 9$, included angle $C = 60^\circ$. Find c to the nearest tenth.

a. 11.5

b. 12.5

c. 10.8

d. 22

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44 [Challenging] • Target: Angle between vectors • Cross-Lesson: 6.01 Limits

In ΔABC, $a = 15$, $b = 7$, included angle $C = 60^\circ$. Find c.

a. 13

b. 14

c. 9

d. 17

45 [Challenging] • Target: Addition and scalar multiplication • Cross-Lesson: 1.01 Complex Numbers

A linear transformation in the 2D plane is represented by the matrix $M = (a, b; c, d)$. This transformation maps the point $P(1, 2)$ to $P'(-3, 7)$ and the point $Q(3, 1)$ to $Q'(5, 1)$. Determine the values of a, b, c, and d, and then calculate the sum $a + b + c + d$.

a. 9

b. 11

c. 7

d. 5

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